

KEVIN YANG

AI / Machine Learning Engineering

NYU M.S. candidate building production AI systems — agent orchestration, retrieval, model post-training, and automated evaluation — on reliable, observable high-throughput backends.

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EDUCATION

New York University — M.S. in Computer Engineering, AI & Data Modeling; GPA 3.8+

Sep 2025 – May 2027

The Chinese University of Hong Kong — B.Eng. in Computer Engineering, Top 20% of program

Sep 2021 – Jun 2025

Selected Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, Computer Vision, Distributed Systems

TECHNICAL SKILLS

Machine Learning & Modeling: PyTorch | LoRA / SFT | DPO | YOLO | OpenCV | TensorRT | Model Evaluation & Benchmarking

AI Systems — Agents, Retrieval & Evaluation: OpenAI SDK | LangGraph | RAG | BM25 | Dense Retrieval | RRF | Cross-Encoder
Reranking | Tracing | Fault Injection

Languages & Backend: Python | Java | SQL | FastAPI | Spring Boot | MyBatis-Plus

Data, Infrastructure & Operations: PostgreSQL | MySQL | Redis | RocketMQ | MinIO | Docker | Kubernetes | Linux | Git | Monitoring & Observability

AI-Assisted Development: AI coding assistants for code review, load-test triage, and defect screening — with explicit verification of model output

EXPERIENCE

Yanfeng Automotive Interiors — AI / Machine Learning Engineer Intern, Corporate Strategy

May 2026 – Aug 2026

VIA — Enterprise decision-intelligence and forecasting platform · Novi, MI

- Owned system architecture and end-to-end delivery, working directly with **plant, finance, commercial, and leadership stakeholders** to turn ambiguous questions on program demand, plant capacity, and 12-month forecast risk into auditable Lead/Data/Forecast agent workflows; built the Python/FastAPI, OpenAI SDK, and LangGraph agent harness with typed task and evidence contracts.
- Engineered a durable agent control plane — PostgreSQL run/event ledgers, leases and fencing tokens, idempotent steps, bounded retry/timeout states, replayable SSE streams — giving resumable investigations and partial-result recovery without duplicate tool side effects.
- Reworked context construction and enterprise retrieval with exact state lookup, bounded cited context, **BM25 + dense retrieval + RRF, and cross-encoder reranking**; fixed a defect exposing only 6 of 24 tool results and improved **R@1 from 0.22 to 0.66 and R@10 from 0.58 to 0.90**, accepting a ~0.9 s p50 budget in exchange for cited, verifiable answers.
- Built **responsible-AI guardrails** into the platform: every answer traces to a cited, versioned record with enforced source, grain, coverage, and provenance, and low-confidence cases route to human review rather than being answered.
- Established **L1–L4 automated evaluations**, tracing, and failure attribution across schema/trajectory invariants and timeout/duplicate/rate-limit fault injection. After finding that **same-model self-judging silently passed over 50% of known failures**, moved versioning, lineage, idempotency, and causal-order checks into deterministic runtime guards instead of trusting LLM judges.

NIO — Computer Vision / Machine Learning Engineer Intern, Advanced Manufacturing

May 2024 – Jul 2024

Production visual-inspection models for a live vehicle assembly line · Shanghai, China

- Post-trained the defect-decision model with **LoRA (r = 16) SFT and DPO** on inspector accept/reject pairs to learn real production tolerance boundaries, raising agreement with senior inspectors from **76% to 91%**.
- Built a governed **YOLO** data and evaluation loop — model-assisted pre-labeling, uncertainty routing for human adjudication, and hidden shadow evaluation to catch evaluator overfitting — raising annotation agreement **71% → 94%** and cutting labeling effort **55%** at matched accuracy.
- Co-deployed the pipeline with **Docker + TensorRT** mixed precision (INT8 backbone, FP16 regression heads) at **~28 ms/frame and 25 FPS**, reaching **≥98% precision and recall** on two defect classes with explicit cross-station drift monitoring.

Capgemini — Backend Software Engineer Intern

Aug 2024 – Sep 2024

Seckill Plus — High-concurrency order, consistency, and workflow platform · Shanghai, China

- Led a modular-monolith + event-driven redesign for flash-sale orders: Redis Lua atomically reserved stock, RocketMQ absorbed bursts, MySQL persisted transaction truth, multi-level caching protected hot keys. Sustained **1,000+ QPS at P99 < 300 ms with 0% errors** across 4,000 burst requests, with zero oversell and zero duplicate valid orders.
- Designed reliable publish and reconciliation with idempotent consumers, durable retries, and compensating checks; across **12 fault-injection scenarios** every anomaly entered a recovery path (messaging < 60 s, cross-store < 4 min). Instrumented **QPS, P95/P99, error-rate, and queue-backlog monitoring with runbooks**.
- Used **AI coding assistants** for code review and load-test analysis, surfacing a Tomcat thread-pool bottleneck and a MyBatis-Plus concurrency bug while screening CPU, memory-leak, and OOM risks — **treating model output as hypotheses to verify under load rather than fixes to apply**.